

DEVANSH JHA

+91 70240-45911 | devanshjha009@gmail.com | Devansh Jha

Summary

Data Engineer with 3 years modernizing legacy ETL into production AWS pipelines, specializing in PySpark, AWS Glue, Step Functions, and Gen AI integration. Build config-driven, multi-format ingestion frameworks, LLM-assisted schema mapping tools, and Step Function–orchestrated workflows that move data from S3 into Db2/Aurora — handling dynamic schemas, API/JSON payload integration, and end-to-end validation. Own the full delivery lifecycle: script development, framework optimization, AI-driven log diagnostics (AWS Bedrock), stored-procedure rework, and automated CI/CD deployment via GitHub Actions & Jenkins, with strong production troubleshooting and observability.

Skills

Cloud & Big Data: AWS (Glue, S3, Lambda, Secrets Manager, Aurora, RDS, SNS, Bedrock), Apache Spark (PySpark, Spark SQL)
AI & Gen AI Integration: AWS Bedrock, RAG (Retrieval-Augmented Generation), Prompt Engineering, LangChain, LLM API Integration, AI Log Diagnostics
ETL & Data Engineering: ETL Pipeline Development, Config-Driven Ingestion, Data Transformation, Data Validation, Schema Mapping, Batch Processing, Data Lineage
Languages: Python, SQL
APIs & Integration: REST API integration, JSON request/response generation, payload mapping & validation
DevOps & CI/CD: GitHub Actions, Jenkins, AWS CloudFormation, Amazon CloudWatch, AutoSys, Git (GitHub, Bitbucket)
Databases: IBM Db2, AWS Aurora, AWS RDS

Work Experience

Cognizant Technology Solutions — Noida, Uttar Pradesh

Programmer Analyst (Sept 2024 – Present)
Programmer Analyst Trainee (Sept 2023 – Sept 2024)

Client: Prudential Financial (Fortune 500) — Enterprise Data Platform Modernization

- Migrated 4+ business-critical legacy Informatica ETL workloads to AWS-native pipelines (Glue, PySpark, S3, Lambda, Step Functions) as part of a multi-job modernization program relied on by multiple downstream teams.
- Engineered a config-driven, multi-format ingestion framework (CSV, XLSX, ZIP, DAT) with schema, delimiter, and table mappings externalized to JSON config — enabling onboarding of new vendor feeds without code changes.
- Developed an LLM-assisted schema mapping utility using AWS Bedrock and Python to analyze dynamic vendor file headers, reducing manual onboarding effort for incoming file feeds.
- Built an AI-powered pipeline diagnostic framework leveraging AWS Lambda and AWS Bedrock to automatically parse Glue/Step Function failure logs, generate root-cause summaries, and send actionable remediation alerts via SNS.
- Built a multi-stage, Step Function–orchestrated Glue architecture for automated claim creation — stage load, batch generation (batches of 150), Lambda-based API invocation, and history load — consumed by several internal teams.
- Integrated the claims REST API via Lambda: generated JSON request payloads from batched records, invoked the API per request, and persisted responses for downstream processing — with partial-success handling to commit only valid API responses.
- Implemented and provisioned database tables across AWS Aurora cluster and RDS instances in Dev and QA environments using automated GitHub Actions workflows.
- Developed a comprehensive data validation framework with row-count reconciliation (source vs. target), schema and data-type checks, null/duplicate detection, and checksum-based integrity verification — with automated SNS alerts on validation failures.
- Optimized PySpark jobs for performance — tuned partitioning and repartitioning strategies, reduced S3 read/write costs, and cut job runtimes from ~30 minutes to 5–6 minutes on high-volume data sets.
- Built Python automation scripts for scheduled file movement, S3 cleanup, monitoring alerts, and data extraction reporting — orchestrated via AutoSys and CloudWatch Events for hands-free operation.
- Reworked IBM Db2 stored procedures to execute directly from AWS Glue jobs (replacing legacy Informatica invocation), diagnosed and resolved procedure-level failures to ensure correct claim processing on AWS.
- Owned CI/CD delivery across GitHub Actions, Jenkins, and CloudBees, led source-control migration from Bitbucket to GitHub, enabling automated, merge-based deployments per environment.
- Managed production incident response and on-call support — triaged pipeline failures, performed root-cause analysis, and deployed hotfixes with zero data loss using CI/CD with no disruption.
- Mentored junior engineers on PySpark best practices, AWS Glue job design, and CI/CD workflows, accelerating their ramp-up and enabling independent delivery.

Projects

Enterprise ETL Modernization Pipeline — Informatica → AWS Migration

AWS Glue (PySpark), Step Functions, Lambda, Amazon S3, IBM Db2, Jenkins/CloudBees, GitHub

Architecture: S3 (multi-vendor inbound) → Glue stage load → Step Functions → Glue batch generation → Lambda (claims API) → Glue claim creation → Db2 (stored procedures) → history load

- Rebuilt a legacy Informatica claim-creation flow as a parameterized, multi-Glue-job pipeline on AWS, depended on by multiple internal teams.
- Designed parameter-driven execution (category + process type) with JSON-configured extraction, delimiters, and filters — handling CSV, XLSX, ZIP, and DAT inputs across multiple vendor feeds.
- Implemented Step Function orchestration for batch processing (groups of 150) with JSON API payload generation, Lambda-based API invocation, and partial-success semantics for mixed API outcomes.
- Delivered a four-stage SNS notification model (initiation, failure, partial success, completion) for proactive pipeline monitoring.

AI-Powered Log Diagnostics & Schema Automation Suite

AWS Bedrock, Python, AWS Lambda, LangChain, Amazon SNS, Amazon S3

- Created an intelligent log analysis utility using Lambda and AWS Bedrock (Claude model) to parse PySpark/Glue error tracebacks and send plain-English root-cause diagnoses to engineers via SNS.
- Implemented an automated LLM schema parser to inspect incoming vendor headers (CSV/XLSX) and generate formatted JSON extraction configurations, speeding up onboarding of complex feed layouts.

Data Validation & Reconciliation Framework

PySpark, AWS Glue, Amazon S3, Amazon SNS, Python

- Designed and built a reusable data quality framework to validate migrated ETL outputs against legacy source systems across all modernized pipelines.
- Implemented row-count reconciliation (source vs. target), schema and data-type validation, null/duplicate detection, and checksum-based integrity checks — catching data discrepancies before downstream consumption.
- Integrated automated SNS alerting on validation failures, reducing manual QA effort and enabling faster incident response for production data issues.

Production Automation & Monitoring Toolkit

Python, AWS CloudWatch, AutoSys, Amazon S3, Amazon SNS

- Built a suite of Python scripts automating file movement, S3 cleanup, and data extraction reporting — replacing manual, error-prone operational tasks.
- Configured CloudWatch Events and AutoSys schedules for hands-free execution of recurring jobs, monitoring scripts, and alerting workflows.
- Reduced operational overhead by automating routine production tasks, freeing engineering time for feature development and migration work.

Education

Bachelor of Technology

Institute of Technology & Management Gwalior — CGPA: 8.49

Aug 2019 – Jun 2023
Gwalior, Madhya Pradesh

Certifications

- * Microsoft Certified: Azure Fundamentals AZ-900 (2025)
- * AWS Certified Cloud Practitioner (2025)